

# Purushottam Kumar

+91-8757422328 | 772purushottam@gmail.com | LinkedIn | GitHub | LeetCode | Portfolio

## EDUCATION

### National Institute of Technology, Patna

*B.Tech, Electronics and Communication Engineering | CGPA: 8.45/10*

Jun. 2024 – May 2028

*Patna, Bihar, India*

## PROJECTS

### Real-Time Transaction Risk Scoring & AML Detection Engine | [Live Demo](#) | [GitHub](#)

- React, Node.js, Express.js, MongoDB (Time Series), Socket.io, Docker | REST API
- Built a real-time transaction monitoring system simulating bank-grade fraud and AML detection, evaluating live transactions against velocity, amount-anomaly, geographic, and structuring risk rules.
- Engineered a transparent, weighted rules engine producing an auditable risk score and decision (auto-approve, flag, or block) with a full explainability breakdown for every flagged transaction, prioritizing regulatory-style determinism over black-box AI scoring.
- Modeled high-velocity transaction data using MongoDB Time Series collections and built a Node.js/Express API to ingest, score, and persist transactions with account-level historical context.
- Implemented a Socket.io real-time pipeline to push flagged transactions instantly to a React dashboard, featuring a live color-coded feed, click-to-expand rule breakdown, and a live risk-score trend chart.
- Containerized the full stack with Docker Compose and deployed it end-to-end – frontend on Vercel, backend on Render, and database on MongoDB Atlas – with unit tests (Jest) covering all core scoring logic.

### Full-Stack URL Shortener | [Live Demo](#) | [GitHub](#)

- React, Node.js, Express.js, REST API
- Built a full-stack URL shortening platform with React, Node.js, and Express, letting users generate, manage, and track custom short links.
- Implemented shortcode validation, duplicate detection, and configurable link expiration on an in-memory data store.
- Exposed a dedicated API endpoint for on-demand QR-code generation, giving every shortened link a scannable QR counterpart.
- Built reusable React components – auth, link creation, analytics, QR display – with client-side validation for a consistent UX across the app.

## ACHIEVEMENTS & CERTIFICATIONS

- Semi-Finalist, Flipkart GRiD 8.0 – advanced past initial screening rounds among 1.7 lakh+ applicants nationwide.
- Solved 1,000+ DSA problems across [LeetCode](#), [CodeChef](#), and [Codeforces](#).
- Secured Elite standing (Top 5%) with a 96% score in NPTEL's "The Joy of Computing using Python," IIT Madras.
- Completed "VLSI SoC Design using Verilog HDL" certification from Maven Silicon, Centre of Excellence in Semicon.
- Achieved 98.6 percentile (All India Rank 22,255) in JEE Main 2024 and qualified JEE Advanced 2024.

## TECHNICAL SKILLS

**Languages:** C++, Java, Python, C

**Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems

**Frontend:** React.js

**Backend:** Node.js, Express.js

**Databases:** MongoDB

**Tools:** Git, GitHub, Docker, IntelliJ IDEA, VS Code